

Swaption model		2	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	0.1273
2	5.110%	5.113%	1x2		d_2	-0.1273
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4		$FN(d_1) - XN(d_2)$	0.005785
5	5.600%	5.635%	4x5		Payer swaption (in bps)	5,785,353 49.5
F	5.712%		$A =$	0.8562		
σ	18.0%					
T	2		Notional	1,167,982,789		
X	5.712%					

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	0.1131
2	5.110%	5.113%	1x2		d_2	-0.1131
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074	$FN(d_1) - XN(d_2)$	0.005383
5	5.600%	5.635%	4x5	0.7602	Payer swaption (in bps)	5,382,623 130.5
F	5.976%		$A =$	2.4238		
σ	16.0%					
T	2		Notional	412,571,222		
X	5.976%					

Figure 8.13: Pricing Up the Swaptions

Figure 8.14 shows the payout profile of the conditional curve trade as a function of where 1-year and 3-year swap rates are at the expiry of the options in two years. Note that both options finishing in-the-money does not necessarily mean that we will have a net positive payout from the options;

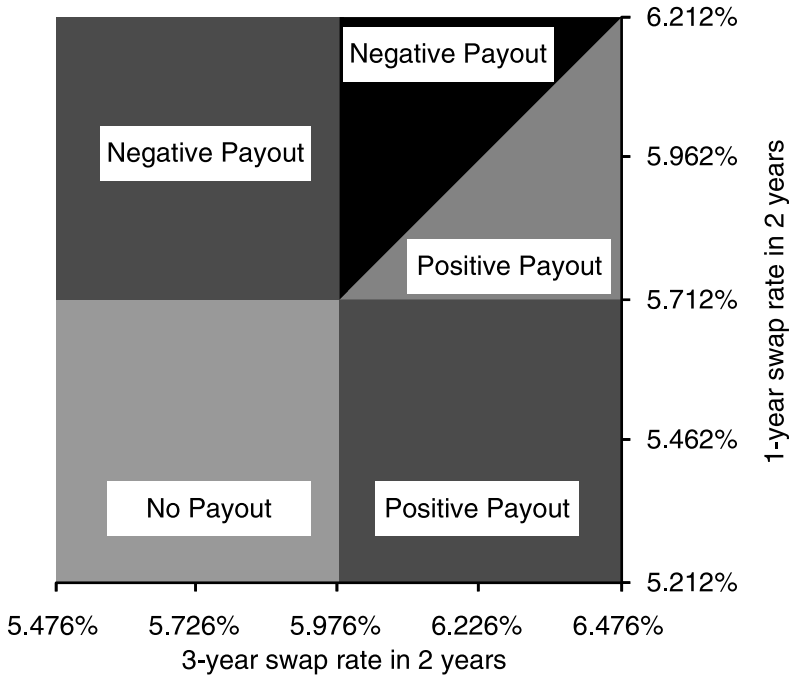


Figure 8.14: Net Payout at Expiry of Option Position (Exclusive of Upfront Fee) as a Function of Then-Prevailing 1-Year and 3-Year Swap Rates

if the curve is steeper than 26.4 bps in this scenario then we will have a positive payout at expiration, and if it is flatter than 26.4 bps, then we will have a negative payout at expiration. Furthermore, it is possible for the curve to steepen dramatically at expiry but for there to be no payout from either of the options. For example, if in two years the 1-year swap rate is 4% and the 3-year swap rate is 5%, we will find ourselves in the lower left-hand quadrant with no payout. Note that it is also possible for exactly one of these options to finish in-the-money. This corresponds to us finishing in either the upper left-hand quadrant or the lower right-hand quadrant.

There are a few other important things to recognize about this trade. First, the upfront amount of \$402,730 that we collect should not be regarded as free money. There is a reason we are getting it. We are selling one option that is more valuable than the option we are buying. In fact, once the upfront fee has cleared and is in our pocket, the mark-to-market on the trade will be $-\$402,730$, all else equal.

Second, this conditional swap curve trade introduces volatility risk. If after entering the trade, 2 into 3 swaption vol cheapens relative to 2 into 1 swaption vol, that will hurt us on a mark-to-market basis. In addition, as rates move around, the strikes on the two options may be significantly away-from-the-money, which means we must take skew into account when pricing the options. These are sources of risk that are very different from the essential view on the curve that we may have started with.²⁸

Another Variation of the Trade

In another variation of this trade, we can get rid of the upfront fee and instead adjust one of the strikes of the swaptions so as to improve our entry level, should the options expire in-the-money. As shown in Figure 8.15, we have solved for the strike on the 2 into 3 payer swaption that we are buying such that it has the same value as the 2 into 1 payer swaption that we are selling.²⁹ In this version of the trade, there is no upfront fee, the strike on the 2 into 1 payer we are selling remains 5.712%, but the strike on the 2 into 3 payer we are buying is now 5.889%. Thus, the strike differential of $5.889\% - 5.712\% = 17.7$ bps affords us a better entry point for the steepener than does the other version of the trade that has an upfront payment but a strike differential of 26.4 bps. Once again, both trades are NPV = 0 transactions. The choice of which one is preferable is up to the individual putting on the trade.

²⁸There are cases in fact in which someone who puts on this trade is correct on the curve component of the trade (i.e., the curve steepens in forward space), but their P&L is overwhelmed (to the downside) by an adverse move in vols. There are some traders who prefer to avoid trades like this, believing if they have a view on the shape of the curve they should not obfuscate that view by comingling it with a trade that has a vol position. Other traders don't mind taking on the vol position for enhanced entry levels, especially when the net result is to provide them with a trade that more closely matches their view. Still other traders might put this trade on with little or no belief that either option will finish in-the-money; they believe that the market will rally and are happy to take the upfront payment from the trade with no expectation of any further payout. In this conditional curve steepener, there is a payoff when the curve steepens conditional on the market selling off. If the market does not sell off, the options will finish out-of-the-money, and it doesn't matter where the curve ends up.

It is also important to note that after we put this trade on, as rates move around we may not remain delta-neutral. The notionals on this trade were chosen so as to provide a delta-neutral trade, and yet with the passage of time and movement of rates we may end up long or short the market. In practice, investors may tweak the notional on either or both legs of the trade over time in order to remain delta-neutral.

²⁹We have assumed for simplicity that changing the strike on the 2 into 3 payer swaption does not change the implied vol used to price the option. That is, we have assumed flat skew. Given the skew prevalent in the swaption market, this assumption may be inaccurate.

Swaption model		2	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	0.1273
2	5.110%	5.113%	1x2		d_2	-0.1273
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4		$FN(d_1) - XN(d_2)$	0.005785
5	5.600%	5.635%	4x5		Payer swaption (in bps)	5,785,353 49.5
F	5.712%		$A =$	0.8562		
σ	18.0%					
T	2		Notional	1,167,982,789		
X	5.712%					

Swaption model		2	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	0.1773
2	5.110%	5.113%	1x2		d_2	-0.0490
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074	$FN(d_1) - XN(d_2)$	0.005785
5	5.600%	5.635%	4x5	0.7602	Payer swaption (in bps)	5,785,353 140.2
F	5.976%		$A =$	2.4238		
σ	16.0%					
T	2		Notional	412,571,222		
X	5.889%					

Figure 8.15: Solving for the Strike on the 2 into 3 Payer Such That It Is Worth the Same as the 2 into 1 Payer

8.3 Trading Swap Spreads

Recall from Section 2.2.1 that the summer of 2003 was a very volatile period of time in the fixed income markets, and this was particularly so for swap spreads. Figure 2.4 displayed the movements in swap spreads and Treasury yields during this time. This was a period of time that was ripe for speculation on the direction of swap spreads. Those who had insight into the dynamics that caused swap spreads to increase dramatically during this time, then only to collapse again toward the end of the summer, could make huge profits if they put on the right trades at the right time.³⁰

As we discussed in Chapter 2, if we are *long swap spreads* or *buy swap spreads*, we are expressing the view that swap spreads increase. If we are *short spreads* or we have *sold spreads*, we are expressing the view that swap spreads decrease.³¹ Figure 8.16 shows an example of a client entering into a long spread position on 5-year swap spreads. As shown in this figure, the client pays fixed in five years and simultaneously buys (a duration-weighted amount of) 5-year Treasuries.³² Oftentimes, the client who puts on a spread position trades the Treasuries with the swap dealer himself. Thus, the swap dealer has on the reverse spread position that the client has put on. From the point of view of the client, the position shown in Figure 8.16 will benefit if 5-year swap spreads widen. Intuitively, if the client has paid fixed at a spread of 21 and a moment later spreads are at 22, that is good for him. By paying fixed and buying Treasuries, he has hedged against the duration position inherent in the swap but remains exposed to changes in swap spreads.

To take a particular example, suppose an investor wants to go long 5-year swap spreads. Refer now to the composite page depicted in Table 1.1. An investor who wants to buy spreads will be paying fixed on the offer side,

³⁰Those who entered into trades too soon that expressed the view that the move in swap spreads was overdone after swaps spreads increased, say, the first 10 bps that summer, were quickly underwater and were often stopped out.

³¹Some would classify a swap spread trade as a macro trade, since it is a trade in which one takes an outright view on a financial variable either going up or going down. However, as will become clear in Section 8.4, it is also correct to view a swap spread as the relative credit spread or liquidity premium between Treasuries and the swap curve, which suggests more of a relative value bent.

³²A Treasury security that matures in exactly five years will typically have a duration that is greater than a spot starting 5-year swap. This would suggest that the duration-weighted number of 5-year Treasuries used to hedge a swap is always less than the notional of the swap itself. However, whereas the maturity of a spot starting 5-year swap is exactly five years, the maturity of the on-the-run 5-year Treasury may be considerably less than five years which, everything else the same, will increase the number of Treasuries needed to make the trade duration-neutral. From a practical point of view, for a 5-year swap with a notional of \$100mm, the duration-weighted number of 5-year Treasuries typically varies between \$97mm and \$103mm.

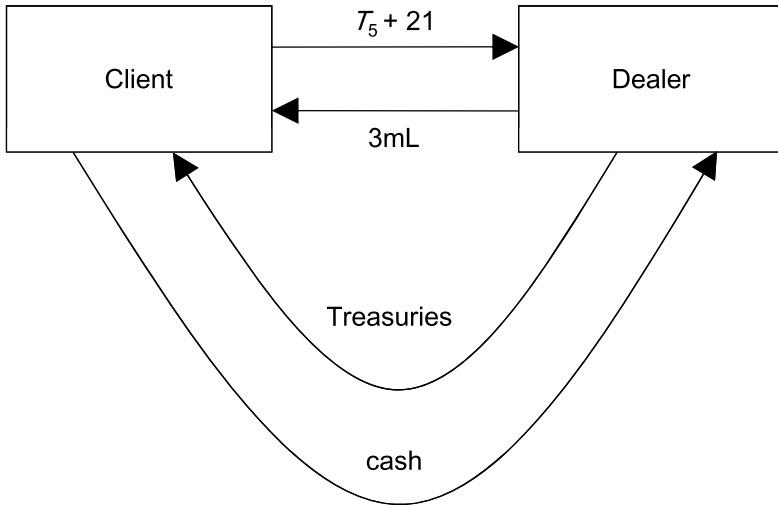


Figure 8.16: Entering in a Long Swap Spread Trade

and thus it is the offer side spread in 5-years of 21 bps that is relevant here. Conversely, if the investor wanted to sell spreads, he would receive fixed on the bid side, and thus the bid side spread of 17 would be where he would get done. Suppose that the investor wants to pay on \$100mm notional of the swap, and further suppose that the duration-weighted number of 5-year Treasuries he should buy is \$98mm. Assume that the investor and swaps dealer agree to exchange the Treasuries at the price of $101-15\frac{3}{4}$, which equates to a yield of 1.438%.³³ Thus, the fixed rate in the swap is $1.438\% + 0.21\% = 1.648\%$.

Suppose that as we approach the end of the trading day, the market for 5-year swap spreads is 22 on the bid side and 26 on the offer side, and the investor decides to unwind his position. In order to unwind, the investor is going to have to receive fixed, and thus it is the bid side of the spread market

³³The exact price at which the two counterparties agree to trade Treasuries is not very crucial in determining the economics of a spread trade. Any price that is agreed on is converted to a yield, which is used to determine the fixed rate in the swap. So if the investor suggests a lower price to sell the Treasuries to the dealer (which equates to a higher yield), a higher yield will be used to determine the swap rate that the investor will be paying. So the lower price at which the Treasuries are sold, although seemingly advantageous, is exactly offset by the higher swap rate that will be paid. Thus, assuming the hedge is delta-neutral, there is no real advantage to be had by trying to alter the Treasury price from one level to another. The market standard is to set the Treasury price on-market, meaning at the price at which the Treasuries are trading at that time.

that is relevant. Suppose that, at the time, 5-year Treasuries are trading at 101-18+ (which is a yield of 1.420%), and the two counterparties agree to exchange Treasuries at that price.

Intuitively, the investor bought spreads at 21, and he is getting out at 22, so we will expect that his P&L will be the present value of 1 bp running on the 5-year swap.³⁴ As a check for the P&L on this trade, we can go ahead and compute the present value of 1 bp running by setting

$$\begin{aligned}\text{PMT} &= 1 \text{ bp}/2 = 0.5 \text{ bp} \\ n &= 10 \\ i &= 1.64\%/2,\end{aligned}$$

which suggests that the P&L on the trade should be around \$48,000. In order to determine his P&L exactly, we need to consider both the gain or loss implicit in the Treasury position as well as the unwind of the swap.

Again, this is a spread trade, and thus its value does not depend on movements in Treasuries. However, it is most appropriate to consider the change in value of each component of the position. Although the 5-year Treasury price has moved, we expect any gain in the Treasury to be offset by a loss in the swap due to the movement in the Treasury component of the rate. The trader bought Treasuries at 101-15 $\frac{3}{4}$ and sold them back later that day at 101-18+. Thus, he has a gain of 2 $\frac{3}{4}$ ticks on his Treasury position, which is worth

$$\$98\text{mm} \times \left(101-18+ - 101-15\frac{3}{4}\right) = \$84,218.75. \quad (8.3)$$

In order to figure out the gain or loss in the swap, recall that the trader had originally paid fixed at 1.648%, and in order to unwind that position, the trader will essentially be receiving fixed at the now-prevailing swap rate of 1.42% + 0.22% = 1.64%. Thus, the swap is out-of-the-money to the trader, and it is as if the trader is paying an annuity of 1.648% - 1.64% = 0.8 bps. In order to find the approximate value of the swap, set

$$\begin{aligned}\text{PMT} &= 0.8 \text{ bp}/2 = 0.4 \text{ bp} \\ n &= 10 \\ i &= 1.64\%/2,\end{aligned}$$

and we solve for PV = 3.8 bps of the notional, or around \$38,000. Thus, the total profit on the trade is the profit from the Treasuries as computed

³⁴In this example, mid-market swap spreads have actually moved 5 bps on the day, but the trade has made only a 1 bp profit. This is due to the 4 bp bid/offer in swap spreads that the investor had to overcome. Oftentimes the bid/offer spread in swap spreads is much tighter than this.

in Equation 8.3, less the loss on the swap, which is $\$84,218.75 - \$38,000 \approx \$46,000$. This is very close to the number \$48,000 that we suggested would be in the ballpark.

8.3.1 Spread Trades for Longer Holding Periods

Thus far we have looked at a spread trade that was unwound the same day that it was put on. Oftentimes people hold spread trades for days, weeks, or even months. The valuation of such trades becomes a little bit more complicated. If the trade is not being unwound the same day that it was put on, then the Treasury position in Figure 8.16 must be financed. The investor will need to enter into a repurchase agreement for these purposes. The valuation of the trade to the investor will incorporate the financing costs that he must bear in order to hold the Treasuries, as well as interest earned on the long Treasury position itself. In addition, once we get beyond the first day of the trade, the investor starts accruing the fixed rate and the floating rate on the swap. Collectively, all of these daily (accrued) cash flows constitute the *carry* of the position.

Depending on the relative level of swap spreads, LIBOR, and the repo rate, the carry may be either positive or negative on the trade. For relatively short holding periods, the carry of the trade is typically of lesser significance than the reason the trade was put on in the first place, namely, to express a view on a move in the underlying swap spread. However, for trades that are held for long periods of time, the carry can definitely add up, having a significant impact on the P&L of the trade.

Of course, in addition to the carry on the trade, we must also consider the mark-to-market of the remaining position. Suppose, for example, that we hold a 5-year spread trade for six months and then decide to unwind it. We will have to consider the carry over the six-month holding period. When it comes time to unwind the trade, it will no longer be appropriate to look at where 5-year swap spreads are trading; six months have gone by and the 5-year spread trade is now a 4.5-year spread trade. That is, our trade has rolled down the curve, and its current valuation is now a function of 4.5-year swaps rates and 4.5-year Treasury yields, not 5-year swap rates and 5-year Treasuries.³⁵

³⁵Unlike carry and roll-down analysis, we must value the rolled-down swap as a function of where the market currently is, as compared to the ex ante analysis in which we assume the swap curve is unchanged between the initiation of the trade and the investment horizon.

Example

Consider once again our stylized example and suppose an investor buys 4-year swap spreads. The investor has the view that swap spreads will increase. Assume the market is as indicated in Table 8.1. In order to proceed with this example, we will make the following simplifying assumptions in our stylized example:

- Treasuries of each yearly maturity are issued annually on the same day at a price of par.
- Today is the day upon which Treasuries have been issued.

When the investor buys \$100mm 4-year spreads at 28, he pays fixed in a \$100mm 4-year swap and buys \$100mm 4-year Treasuries at a price of par.³⁶ Note that since the 4-year Treasury has been issued today at a price of par, its coupon is 5.19%. Assume the investor plans to hold this position for a year and finances the Treasuries by entering into a term repo agreement at a rate of 4.80%. Furthermore, assume there is no haircut. Suppose that a year later the market is as presented in Table 8.9. Let's look at the P&L on the trade. The carry on the trade over the year is composed of the coupon earned on the Treasury, the cost of financing the Treasury, and the net cash flow on the swap. The coupon interest earned on the Treasury is

$$\$100\text{mm} \times 5.19\% = \$5,190,000.$$

The cost of financing the Treasuries is

$$\$100\text{mm} \times 4.80\% = \$4,800,000.$$

And the net cash flow on the swap is

$$\$100\text{mm} \times (5.00\% - 5.47\%) = -\$470,000.$$

Thus, the net carry on the trade for the year is $\$5,190,000 - \$4,800,000 - \$470,000 = -\$80,000$.

We now need to consider the mark-to-market on the outstanding position in a year. In fact, let's suppose that the investor decides to unwind the spread trade at this time. Thus, he will sell the Treasuries and unwind the swap. The Treasuries that were bought one year ago at par now have three years remaining and are yielding 5.19%. Since this is the coupon on

³⁶Based on the different PV01s of the swap and the Treasury, a duration-weighted Treasury hedge would consist of \$99.73mm Treasuries. We will just round this to \$100mm Treasuries.

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
1	5.19%	27	5.46%
2	5.19%	27	5.46%
3	5.19%	27	5.46%
4	5.19%	27	5.46%

Table 8.9: The Market in One Year

the bond, this means that the Treasuries are trading at par.³⁷ Thus, there will be no P&L associated with the sale since they were purchased at this price (i.e., the sale of the Treasuries at par will provide the proceeds needed to offset the funds needed to close out the repo transaction).³⁸

Now let's consider an unwind of the swap. The investor is currently paying 5.47% in a swap with three years remaining. The 3-year swap rate is 5.46%. It can be shown that the mark-to-market on this trade is $-\$26,999$. Thus, the total P&L on the trade is

$$-\$80,000 - \$26,999 = -\$106,999.$$

In this case, the negative carry on the trade (which was $-\$80,000$) was a major contributing factor to the P&L loss on the trade.³⁹ $\square$

8.3.2 Spread of Spread Trades

Sometimes people put on spread trades that do not involve the outright movement in a single swap spread but rather the relative movement of one swap spread to another. Such a trade is referred to as a *spread of spreads* trade. Figure 8.17 shows a graph of both 10-year and 2-year swap spreads

³⁷We are making the implicit assumption here that there is no premium for on-the-run Treasury securities. In a year, the 4-year Treasuries that we had purchased at the inception of the trade are now 3-year off-the-run Treasuries with a coupon of 5.19%. And a newly issued 3-year Treasury issued at par also has a yield of 5.19%.

³⁸The total P&L on the Treasury should incorporate the financing charge incurred on the Treasuries.

³⁹In this trade, the investor bought 4-year spreads at 28, held the position for a year, and took a loss when spreads finished at 27. How much would spreads have had to widen in order for the investor to break even?

When the investor pays fixed in a 4-year swap today and holds the position for a year, he will experience negative carry of $5.00\% - 5.47\% = 47$ bps. We have previously shown that the 1x4 swap rate given the curve in Table 8.1 is 5.644%. That is, the market expects that the 3-year swap rate in one year will be 5.644%. Although we are experiencing negative carry in the first year, the expected mark-to-market on the residual 3-year swap with a